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MASTER THESIS

**The Impact of COVID-19 Restrictions on the Employment of Workers with
Disabilities**

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Introduction

The COVID-19 pandemic triggered an unprecedented contraction in labor markets worldwide, characterised by widespread job losses, reduced working hours, and declining earnings. While the aggregate effects of the pandemic on employment and wages have been extensively documented, a growing body of research reveals that these adverse consequences were not distributed uniformly across the working population. Rather, pre-existing structural inequalities based on gender, race, age, and disability status were systematically amplified by the crisis with significant magnitude. Among these axes of stratification, disability status remains notably under-researched in the causal econometric literature about Russia. Despite longstanding evidence that disabled workers face persistent employment and wage gaps under normal economic conditions, relatively little is known about how these disparities evolve during exogenous labour demand shocks.

Russia combines a historically high disability prevalence rate with a labor market characterised by pronounced segmentation, relatively weak social protection mechanisms for disabled workers, and a pandemic response that initially prioritised general economic stabilisation measures over disability-specific safeguards. However, existing studies on pandemic labour market outcomes in Russia have largely focused on aggregate employment trends, gender disparities, or regional variation, leaving the specific experience of disabled workers largely unexplored.

The importance of disability-specific safeguards during black swan events such as the COVID-19 pandemic cannot be overstated. During catastrophic events, marginalized groups become more vulnerable to institutional discrimination and systemic neglect¹. Healthcare systems become overwhelmed, social services are

¹ Kuper, H., Banks, L. M., Bright, T., Davey, C., & Shakespeare, T. (2020). Disability-inclusive COVID-19 response: What it is, why it is important and what we can learn from the United Kingdom's response. Wellcome Open Research, 5, 79. <https://doi.org/10.12688/wellcomeopenres.15833.1>

disrupted, and labour markets contract often with the most severe consequences for those already facing structural disadvantages. For persons with disabilities, exogenous shocks create heightened health risks, disruption of essential support services, and disproportionate labour market exclusion². Without targeted safeguards, the pre-existing employment and earnings gaps between disabled and non-disabled workers can widen dramatically, potentially creating permanent scarring effects that persist long after the immediate crisis has subsided. Understanding whether, how, and to what extent disability-based disparities evolve during such shocks is therefore critical for designing effective policy interventions that prevent crisis periods from entrenching long-term labour market inequalities

Russia has well established social policy regarding measures of social protection for people with disabilities. That includes financial support which is based on combination of federal and regional measures. Individuals with disabilities are entitled to state pensions, which vary based on the severity of the disability and work history, including social pensions for those without sufficient employment records (*Federal Law No. 166-FZ of December 15, 2001, “On State Pension Provision in the Russian Federation”³*). In addition, they receive monthly cash payments, which may include both monetary allowances and in-kind benefits such as free medication, transport discounts, and access to sanatorium treatment (*Federal Law No. 178-FZ of July 17, 1999, “On State Social Assistance”⁴*). Many regions also offer supplemental support, including housing subsidies, one-time payments, and tax exemptions. Families raising children with disabilities receive additional monthly allowances and benefits to ease the financial burden of care (*Federal Law No. 181-FZ of November*

² United Nations. (2020). Policy brief: A disability-inclusive response to COVID-19. United Nations.

³ Federal Law No. 166-FZ of December 15, 2001, "On State Pension Provision in the Russian Federation"

⁴ Federal Law No. 178-FZ of July 17, 1999, "On State Social Assistance"

24, 1995, “*On Social Protection of Disabled Persons in the Russian Federation*”⁵). Medical and social rehabilitation where individuals with disabilities are entitled to a comprehensive rehabilitation program tailored to their specific needs (*Federal Law No. 181-FZ “On Social Protection of Disabled Persons in the Russian Federation”*). Employment protection and the quota system has been given a special attention under *Federal Law No. 181-FZ*. Employers are required to create suitable working conditions and ensure equal opportunities for individuals with disabilities. One of the main mechanisms supporting this is the **employment quota system**, which obligates companies with more than 100 employees to reserve 2–4% of jobs specifically for people with disabilities, with regional authorities setting the exact quota within this range. Employers who fail to comply may face administrative penalties. To support implementation, the state offers help on workplace adaptations, vocational training, and subsidies for hiring and retaining disabled employees. But as highlighted by many researchers⁶, this system has many drawbacks regarding quota-based employment system. While the system mandates that companies with a certain number of employees must allocate a percentage of jobs to people with disabilities, compliance is often superficial. Many employers either pay fines or create symbolic positions without genuine responsibilities, undermining the goal of integration. In some cases, quota-based system is not working altogether. Additionally, the quota system does not address broader issues such as workplace accessibility, stigma, or the mismatch between job requirements and the qualifications of disabled individuals. As a result, the system can inadvertently

⁵ Resolution of the Government of the Russian Federation No. 890 of July 30, 1994, "On State Support for the Development of the Medical Industry and Improving the Provision of Medicines and Medical Devices to the Population and Healthcare Institutions"

⁶ Rasell, M., & Iarskaia-Smirnova, E. (Eds.). (2013). *Disability in Eastern Europe and the Former Soviet Union: History, policy and everyday life* (1st ed.). Routledge. <https://doi.org/10.4324/9781315866932>

reinforce marginalization rather than fostering meaningful employment opportunities and social inclusion.

The aim of this research is to evaluate the impact of COVID-19 restrictions on the employment outcomes of workers with disabilities in Russia

The following tasks were set in the course of the study:

- 1) Systematize theoretical approaches to the analysis of labour market performance of people with disabilities and different types of discrimination frameworks.
- 2) Identify key measurements of employment outcomes that reflect the impact of pandemic restrictions on employment and key variables that influence disability as an acquired status.
- 3) Conduct a quantitative analysis using a two-way fixed effects (TWFE) difference-in-differences (DiD) event study model to evaluate the impact of pandemic restrictions on labor outcomes.
- 4) Formulate practical recommendations for regulatory bodies to mitigate the effects of shock events on workers with disabilities.

The *object* of the study is workers with disabilities.

The *subject* of the study is the impact of COVID-19 restrictions on employment of workers with disabilities.

The following research questions have been formulated:

- 1) What is the causal effect of COVID-19 pandemic restrictions on the monthly earnings of workers with disabilities relative to non-disabled workers in Russia?
- 2) Are the pandemic-induced disparities in labor market outcomes between disabled and non-disabled workers transitory (reversing as the economy

recovers) or persistent (creating a permanent divergence in employment and earnings trajectories)?

The following hypotheses were put forward:

- 1) H1. Disabled workers experience a larger decline in monthly earnings during the COVID-19 pandemic period compared to non-disabled workers.
- 2) H2. The earnings penalty experienced by disabled workers persists beyond the immediate onset of the pandemic (2020) into subsequent years (2021–2022), indicating that the shock produces a medium-term divergence in earnings trajectories rather than a temporary dip.

The first section of this study explores concepts and definitions of disability with measurement approaches linked to disability definitions and the theoretical frameworks that explain the impact of pandemic event on employment earnings of disabled people.

The second section examines empirical studies to understand the relationship between employment outcomes, shock events and socio-demographic factors. Socio-demographic and environmental variables examined that may influence disability outcome. Following part an econometric model is being developed that analyzes the impact of COVID-19 on average monthly earnings of disabled people compared to non-disabled. Propensity scores applied to address non-randomness of the treatment. Dynamic event study applied to assess magnitude of the pandemic effect of average monthly earnings.

Based on empirical findings the final section interprets the results of the model and offers methodological and policy recommendations to address presented challenges in future shock events to mitigate the consequences for disabled people.

The structure of the final thesis includes an introduction, 3 chapters, and a conclusion. The total number of pages is ## pages. The total number of literature sources is ##.

1. Disability, Work, and Structural Inequality: A Theoretical Examination

1.1 Disability definition and measurement

Disability is a wide-ranging concept that refers to either physical, mental, or cognitive impairments arising from a congenital disorder, accident, disease, or the ageing process. Disability refers to a long process resulting from the interaction between an individual's health and various factors, including gender, age, habits, and work. Thus, disability can be seen as the result of a progressive health capital depreciation, an impairment or a nonintegration of the individual in the society⁹.

As of recent research five main approaches of defining the disability had been developed and in use.

Medical model

The medical model views disability as an individual problem resulting from impairments, diseases, or health conditions that require treatment or rehabilitation. The approach focuses on diagnosing and curing disabilities, often overlooking societal barriers and often criticized for neglecting social and environmental factors. This approach is often regarded as the main reason why the social protection model in Russia often is viewed as outdated⁷. The medical model conceptualises disability as a direct consequence of an individual's clinically diagnosed physical, sensory, or cognitive impairment, positing a causal chain from underlying pathology to functional limitation and, ultimately, to social and economic disadvantage⁸. In socio-economic studies, this operationalisation often translates into binary variables derived from clinical screening questions, medical records, or administrative classifications such as receiving a disability pension based on a certified loss of

⁷ Sukushina, E. V., Marinenko, Y. S., & Vyalova, V. P. (2024). Specifics of representing disability in scientific discourse: The experience of a practical study [Особенности представления инвалидности в научном дискурсе]. *Vestnik Tomsk State University. Philosophy. Sociology. Political Science*, (Issue 82), 183–198. <https://doi.org/10.17223/1998863X/82/16>

⁸ Ramaswamy, K. (2024). *Crippling Justice: Enabling Theory on Disability* [Apollo - University of Cambridge Repository]. <https://doi.org/10.17863/CAM.115138>

working capacity (e. g. section about health evaluation in RLMS panel)⁹. The central analytical implication of this approach is its "atomistic individualism": it frames the labour market disadvantage as originating endogenously within the person's functional lack of capabilities, which is assumed to inherently reduce productivity. This definition steers the analytical focus towards supply-side explanations, such as a human capital deficit caused by the impairment, and naturally propagates policy solutions aimed at correcting the individual through medical rehabilitation or compensating for their unproductive status through welfare transfers. Critics argue that by treating disability as a stable, decontextualised individual trait, this measurement approach systematically obscures the role of exogenous discriminatory social environmental barriers, rendering the socio-structural production of disability analytically invisible to researcher¹⁰.

Social model

Developed by disability activists and scholars, the social model argues that disability arises from societal barriers rather than individual impairments¹¹. Approach stands that the disability is caused by inaccessible environments, discrimination, and social exclusion. And distinguished between disability and impairment. While first is caused by societal injustice the second focuses on physical aspect. This reframing established what Oliver described as "the big idea," the recognition that disability is not a personal tragedy but rather a failure of societal design. The model moves the locus of the problem from the individual's functional limitations to the physical, attitudinal, and institutional barriers that systematically exclude them.

Human Rights Model

⁹ RLMS official page (<https://www.hse.ru/en/rlms/questionnaires>)

¹⁰ Pryma, J. (2025). Trauma as a workaround: Recognizing chronic pain as disability without medical documentation in the United States and France. *Social Science & Medicine*, 382, Article 118364.

¹¹ Shakespeare, T. (2006). *Disability rights and wrongs*. Routledge.

This model frames disability as an issue of equality and non-discrimination¹². The model emphasizes on acknowledgment of social barriers and ways to remove them. By definition the model is a social one, regarding disability as impairment rather than a medical diagnosis. This model recognises that disability is a social construct and that impairments must not serve as a legitimate ground for the denial or restriction of human rights.

Biopsychosocial Model

A compilation of all previous models which integrates biological, psychological, and social factors and considers bodily functions, activities, participation, and environmental influences¹³. Originating from George Engel's broader critique of biomedical reductionism, the model was operationalised for disability through the World Health Organization's International Classification of Functioning, Disability and Health (ICF), endorsed in 2001¹⁴. The ICF framework conceptualises disability not as an individual attribute but as the outcome of dynamic interactions between three interconnected domains: impairments at the level of body functions and structures, activity limitations at the personal level, and participation restrictions within broader social contexts, all modulated by environmental and personal factors. This structure deliberately integrates the medical model's attention to underlying pathology with the social model's insistence on environmental barriers, positioning disability as a “relational condition” that can potentially disadvantage individuals within specific environments. The model thus reframes the central question from why an individual cannot work due to their impairment to why someone with a health condition does not achieve expected vocational outcomes, directing attention

¹² United Nations. (2006). Convention on the Rights of Persons with Disabilities (CRPD).

¹³ World Health Organization (WHO). (2021). International Classification of Functioning, Disability and Health (ICF). <https://www.who.int/classifications/icf/en/>

¹⁴ Dan, B. (2024), The ICF as a socio-psycho-biological model for the full participation of disabled individuals. *Dev Med Child Neurol*, 66: 1398-1399. <https://doi.org/10.1111/dmcn.16044>

toward modifiable factors in the workplace, the benefit system, and personal coping strategies.

Critical Disability Studies

An approach with heavily criticism of traditional models, emphasizing the intersectionality of disability with race, gender, and class¹⁵. The model challenges the concept all previous definitions by challenging the concept of normality itself. The field reframes disability not as a problem of exclusion from existing social and economic structures, but as a generative site of critical knowledge that exposes the instability, violence, and contingency of all claims to normalcy.

Regarding disability definitions social model is most appropriate in context of econometrical study. Thus, shifting focus of influential factors from results of medical examinations to discrimination caused by individual's impairments and their perceptions by society.

Disability measurement is not a straightforward process. Two types of disability measures can be used: self-reported and objective assessments.

Self-reported disability status is reported by the respondent and may be biased due to possibility of self-declaration. Examples of such indicators may be GALI (Globally Activity Limitation Indicator) and ADL (Activities of Daily Living)¹⁶.

Objective indicators, such as physical measures, are often used to avoid self-declaration bias and the interpersonal comparison of self-perception problems of self-reported indicators by aiming to have a more accurate measure. It involves direct testing of physical capabilities rather than reported status by the respondent. Physical health measures can be used, such as grip strength and the ability to get up from a chair. To objective measures we can also assign agencies' assessments, as an

¹⁵ Goodley, D. (2017). *Disability studies: An interdisciplinary introduction* (2nd ed.). Sage.

¹⁶ Van Oyen, H., Van der Heyden, J., Perenboom, R., & Jagger, C. (2006). Monitoring population disability: evaluation of a new Global Activity Limitation Indicator (GALI). *Sozial- und Präventivmedizin*, 51(3), 153-161. <https://doi.org/10.1007/s00038-006-0035-y>

example Russian social-medical expertise committee. Even if they may be gained through the results are reported by person directly, the result of assessment is not biased by the respondent.

Measuring disability is not an intuitive process. The assessment may be biased due to self-report error or inadequately selected disability framework. The most reliable in case of socio-econometric studies is social model paired with objective indicators which grants researcher both power selecting social influential factors and objective measure of disabilities.

1.2 Theoretical frameworks regarding disability and labor market

Definition of the disability is closely related to theoretical framework. Theoretical framework provides a basis for analysis of the connection between disability and labor market. This knowledge is crucial to regulatory bodies to provide equal opportunity environment where marginalized individuals can fulfil their potential regardless of their impairments. Generally, there are several approaches to examine influential factors of employment. These approaches can be categorized into the following broad groups: discrimination theories, human capital theories and capability approach.

Human capital theory (HCT)

Developed by economists Gary Becker (1964) and Theodore Schultz (1961), this theory posits that individuals invest in education, skills, and health to enhance their productivity and earnings over time.

According to Becker, investments in human capital can be understood as analogous to investments in physical capital¹⁷. He delineates several critical

¹⁷ Becker, G. S. (1964). Human capital: A theoretical and empirical analysis, with special reference to education. National Bureau of Economic Research; distributed by Columbia University Press.

dimensions: first, individuals determine the extent of their investment in education and health based on anticipated returns; second, an individual's level of education and qualifications directly influences their earnings and economic participation. Becker further highlights that disparities in human capital endowments across population groups can generate inequalities in both income distribution and access to opportunities.

For workers with disabilities, these investments are often hindered by structural barriers, such as limited access to training and workplace accommodations¹⁸. The pandemic exacerbated these challenges, as lockdowns and remote work transitions disrupted traditional employment pathways, disproportionately affecting vulnerable groups¹⁹. Econometric models, incorporating human capital variables such as education and work experience, can help quantify the pandemic's differential impact on workers with disabilities compared to their non-disabled people.

In terms of human capital disability can be conceptualized as a factor that may influence both the costs and expected returns of human capital investment. For many disabled individuals, investments in education and vocational training are often accompanied by additional expenses, such as accessible learning materials, assistive technologies, transportation expenses, or specialized healthcare. Moreover, anticipated returns on such investments may be systematically lower due to labor market discrimination, physical and social barriers in workplaces, or a lack of reasonable accommodations. Becker himself acknowledged that employer discrimination, based on taste or perceived productivity differences, could distort market outcomes. Consequently, even when disabled individuals possess equal or

¹⁸ Haveman, R., & Wolfe, B. (2000). The economics of disability and disability policy. *Handbook of Health Economics*, 1, 995-1051

¹⁹ Jones, M. K., Latreille, P. L., & Sloane, P. J. (2021). Disability, gender, and the British labour market. *Oxford Economic Papers*, 73(2), 565-587.

higher levels of formal human capital compared to non-disable, their employment rates and earnings frequently lag behind – a phenomenon that signals a market failure in the efficient allocation of human capital.

Health capital also plays crucial role. Health investments, including rehabilitation, assistive devices, and ongoing medical care, directly affect the productive capacity and labor supply of disabled people. However, if these investments are underfunded or inaccessible, the formation of human capital among disabled individuals is compromised from an early age. Furthermore, employers may underestimate the productivity of disabled workers due to inaccurate perceptions of health-related limitations, thereby reducing the incentives for disabled individuals to invest in further education or job-specific skills²⁰. This creates a suboptimal equilibrium: lower demand for labor leads to lower expected returns, which in turn reduces investment incentives, perpetuating lower employment and earnings.

In general, HCT allows provides a robust analytical framework for understanding the employment challenges faced by disabled people. Also this approach fits discreetly with social definition of disability. However the HCT does not take into account external policy changes or event related factors such as pandemic.

Demand for health capital

The concept of health capital, introduced by Michael Grossman in 1972, extends the principles of human capital theory to the domain of healthcare, framing health as an investment good that individuals "produce" through medical care, lifestyle choices, and environmental factors. Unlike traditional consumer goods, health capital is unique because it directly influences an individual's productive capacity, longevity, and overall quality of life. The demand for health capital is

²⁰ Gilleskie, D. B., & Hoffman, D. (2014). Health capital and human capital as explanations for health-related wage disparities. *Journal of Human Capital*, 8(3), 235–279

shaped by a person's desire to maintain well-being over their lifetime, balancing immediate consumption against future health benefits²¹.

Grossman's model posits that individuals act as both producers and consumers of health, investing in medical services, nutrition, and exercise to "depreciate" health capital more slowly. This framework helps explain why higher-income individuals often exhibit better health outcomes – they can afford greater investments in healthcare, healthier diets, and safer living conditions²². However, the demand for health capital is not purely rational or uniform across populations. Behavioral economics highlights that time inconsistency and present bias often lead individuals to underinvest in preventive care (e.g., skipping vaccinations and routine check-ups) despite long-term benefits²³. Moreover, those with higher education tend to demand more health capital because they are more efficient producers of health, lowering the effective cost of health investments. From the perspective of disabled people, this framework implies that any condition accelerating health depreciation or raising the cost of health investment (e.g., inaccessible healthcare or expensive assistive devices) reduces the optimal health capital stock, thereby diminishing labor supply, earnings potential, and overall economic participation.

Critics argue that it underestimates socio-structural determinants of health, such as discrimination, environmental racism, or unequal access to medical innovation²⁴. Newer econometric approaches, such as structural equation modeling,

²¹ Grossman, M. (1972). On the concept of health capital and the demand for health. *Journal of Political Economy*, 80(2), 223–255.

²² Galama, T., & Van Kippersluis, H. (2019). A theory of socioeconomic disparities in health. *Journal of Health Economics*, 62, 1–14.

²³ Baicker, K., Mullainathan, S., & Schwartzstein, J. (2015). Behavioral hazard in health insurance. *Quarterly Journal of Economics*, 130(4), 1623–1667.

²⁴ Williams, D., & Mohammed, S. (2013). Racism and health I: Pathways and evidence. *American Behavioral Scientist*, 57(8), 1152–1173.

attempt to integrate these systemic factors by analyzing pathways like stress-induced biological wear-and-tear²⁵.

In summary, DHC approach is more sensible to pandemic related restrictions hence it studies health capital primely. However as with previous framework this approach has its limitations regarding policy interventions.

Capability Approach

Developed by Nobel laureate Amartya Sen and philosopher Martha Nussbaum, the Capability Approach represents a paradigm shift in welfare economics by evaluating human development not through income or utility alone, but through people's real opportunities to achieve valued ways of living²⁶. This framework distinguishes between functionings (what people actually do and are) and capabilities (the genuine freedoms they must choose among possible functionings). Unlike traditional metrics such as GDP, the Capability Approach provides a multidimensional lens to assess crises, asking not just about survival rates but whether people can live with dignity during and after emergencies²⁷.

Central to the approach is the concept of conversion factors – how individuals transform resources into capabilities based on personal, social, and environmental circumstances. Nussbaum's ten central capabilities: including life, bodily health, and political participation offer a normative framework for policy evaluation. When applied to school closures, this reveals how digital divides not only affected education (a functioning) but diminished future capabilities like labor market

²⁵ Adda, J., Banks, J., & Von Gaudecker, H. M. (2003). The impact of income shocks on health: Evidence from cohort data. *Journal of the European Economic Association*, 1(4), 968–1000.

²⁶ Sen, A. (1999). *Development as freedom*. Oxford University Press

²⁷ Nussbaum, M. C. (2006). *Frontiers of justice: Disability, nationality, species membership*. The Belknap Press of Harvard University Press.

participation²⁸. Structural equation modeling demonstrates these intergenerational capability deprivations particularly impacted girls in low-income countries²⁹.

The approach's emphasis on agency freedom challenges conventional policy tradeoffs. Sen's analysis of famines shows starvation occurs not from absolute food shortages but entitlement failures: an insight applicable to pandemic medical shortages. Countries following capability-oriented policies prioritized both ICU beds (health functioning) and fair allocation systems (health agency), outperforming nations focused solely on medical outputs. Randomized evaluations of vaccine distribution confirm that participatory approaches: involving communities in priority-setting increased both uptake and perceived fairness. Such findings underscore the approach's core thesis: development must expand people's ability to shape their lives, not just deliver goods.

This directly addresses a limitation of both Becker's and Grossman's human capital frameworks, which assume relatively homogeneous conversion capacities. For disabled people, the Capability Approach insists that equality of resources (e.g., same income or education as non-disabled peers) is insufficient for achieving genuine equality of capabilities. Two persons with identical financial resources may have vastly different real freedoms: one with a mobility impairment requires accessible transport and ramps to achieve the functioning of moving freely, whereas a non-disabled person does not. Therefore, justice demands not merely equal inputs but equal capabilities, meaning targeted accommodations and structural changes to remove barriers to conversion.

In summary, this approach provides an alternative to human capital and health capital frameworks by centering analysis on capabilities rather than resources, utility, or productivity. It defines well-being as the ability to achieve valuable

²⁸ Terzi, L. (2020). Capability and educational equality. *Journal of Human Development*, 21(2), 123-139.

²⁹ Kabeer, N. (2021). Women's empowerment and economic development. *World Development*, 147, 105553.

functionings, such as working, participating in community life, and maintaining health, while explicitly accounting for human diversity and conversion factors. For disabled people, the approach is particularly powerful because it reveals that equal resources cannot guarantee equal opportunities when personal, social, and environmental barriers exist. In the employment domain, the Capability Approach advocates for systemic removal of disabling barriers, reasonable accommodations, and respect for diverse work-related choices, rather than narrowly focusing on earnings or labor market outputs.

Labor Market Segmentation Theory

Developed by Peter Doeringer and Michael Piore in their 1971 work *Internal Labor Markets and Manpower Analysis*³⁰, LMST divides the economy into a "primary" sector (characterized by high wages, job security, employer-provided benefits, and internal career ladders) and a "secondary" sector (defined by low wages, high turnover, instability, and few advancement opportunities). LMST argues that labor markets are institutionally divided into distinct segments characterized by fundamentally different rules, incentives, and mobility patterns. These segments are not merely reflections of individual choices or skill endowments but are structured by economic, social, and historical forces, including industrial organization, collective bargaining arrangements, and patterns of discrimination. The theory posits that workers' positions within this segmented structure determine their employment outcomes as much as, if not more than, their personal human capital levels.

The primary sector consists of jobs offering high wages, employment stability, opportunities for advancement, due process protections, fringe benefits, and positive working conditions. These jobs are often located in core industries (e.g.,

³⁰ Doeringer, P. B., & Piore, M. J. (1971). *Internal labor markets and manpower analysis*. Heath

manufacturing, finance, public administration) and are typically filled through internal labour markets, where hiring occurs from within the organization and wages are attached to job positions rather than to individual productivity. In contrast, the secondary sector comprises low-wage, precarious employment characterized by high turnover, limited or no promotion prospects, few benefits, arbitrary supervision, and minimal legal or collective protections. Secondary sector jobs are concentrated in peripheral industries (e.g., retail, hospitality, temporary agencies, domestic work) and operate under external labour markets, where wages are determined by spot market competition.

This theory is very relevant to disabled worker's employment. Disabled workers are overrepresented in secondary sector employment—part-time, temporary, low-wage, and precarious work—where job insecurity is high and workplace accommodations are rarely provided. The segmentation is reinforced by disability benefit systems that create "inactivity traps": leaving secondary sector employment for primary sector jobs is structurally difficult, while taking secondary sector work may jeopardise essential benefits. Internal labour markets in the primary sector often use standardised entry requirements (e.g., physical fitness tests, certain educational credentials obtained within normative timeframes) that implicitly exclude many disabled people without any job-relevant justification. Fourth, the theory predicts that even when disabled workers enter primary sector employment, they may face a "secondary segment within the primary sector"—lower-tier positions with fewer advancement opportunities—a phenomenon sometimes termed internal segmentation.

In summary, by dividing labor market into two sectors characterised by distinct wages, stability, advancement opportunities, and mobility rules the theory highlights barriers between disadvantaged groups such as disabled people and regular workers. Even when entering primary sector employment, they often face

internal segmentation. This framework is highly adaptable both with social definition of disability and policy evaluation strategies.

Institutional Ableism

Developed primarily by Fiona Kumari Campbell in her seminal 2009 monograph, *Contours of Ableism: The Production of Disability and Abledness*, and further developed it in subsequent articles such as “Inciting Legal Fictions: Disability’s Date with Ontology and the Ableist Body of the Law” (2001) and “Exploring Internalized Ableism Using Critical Race Theory” (2008). Drawing from poststructuralist and critical race traditions, Campbell moved beyond individual or attitudinal models of prejudice to expose how ableism operates as an entrenched, often invisible, logic embedded within the rules, practices, and norms of modern institutions. For Campbell, institutional ableism is not merely a collection of discriminatory acts but a deep-seated ontological framework that privileges “abledness” as the natural, desirable, and neutral standard of human being, thereby rendering disability as a deficit, a deviation, or a state of incompleteness³¹.

Central to Campbell’s theory is the distinction between *personal* ableism—overt prejudice by individuals—and *institutional* ableism, which she defines as the systematic production of disability through governance, architecture, policy, and professional epistemologies. She argues that institutions such as education, healthcare, law, and employment operate on unexamined assumptions about human functionality, time, productivity, and spatial organization that systematically exclude or marginalize disabled people. A key aspect is what Campbell terms the “tyranny of the normative”—the relentless demand that bodies and minds conform to statistical averages or idealized capacities. This normativity is not neutral but actively constructed through institutional routines: standardized testing, nine-to-five work schedules, linear career trajectories, and architectural designs that presume

³¹ Campbell, F. K. (2009). *Contours of ableism: The production of disability and abledness*. Palgrave Macmillan.

bipedal mobility and sensory typicality. Moreover, Campbell emphasizes that institutional ableism is often enacted with benevolent intent, masked by discourses of “reasonable accommodation” or “special needs,” which paradoxically reinforce the disabled person’s status as an exception to an otherwise unmarked, able-bodied rule.

Another foundational component of Campbell’s approach is the concept of *internalized ableism*, which she links directly to institutional structures. While internalized ableism refers to disabled people’s own adoption of dominant norms as a measure of self-worth, Campbell insists that this psychological phenomenon cannot be separated from the institutional conditions that constantly communicate inferiority. For example, welfare-to-work programs, medical rehabilitation protocols, and disability benefit assessments often require disabled individuals to perform “recovery,” “effort,” or “compliance” in ways that deny their embodied realities. Campbell calls this the “economy of ableist recognition,” where institutions grant limited access and rights only to those disabled people who approximate able-bodied norms—a process she likens to racial passing or other kinds of assimilation. Thus, institutional ableism for Campbell is a total social fact: it shapes legal definitions of disability, structures professional training, determines funding priorities, and even influences which forms of disabled knowledge are deemed credible within academic and policy circles.

In summary, this approach is a very radical implementation of social model of disability. It sees impairments not only as barriers of disabled workers’ self-realization but also challenges view of institutions like healthcare system as facilitators. By assigning a person status of being disabled the institution itself stigmatizes workers with disabilities.

Comparison of theoretical approaches

To determine the most effective theoretical approach that explains the broadest impact of monetary policy on economic growth, the method of comparative alternatives will be used:

$$U_j = \sum_{i=1}^n m_i u_{ij} \quad (1.1)$$

Where:

n – is a number of criterions;

U_j – is a generalized estimation of j^{th} alternative according to all criterions;

m – is a weight of i^{th} criterion

u_{ij} – is an estimation of j^{th} alternative according to i^{th} criterion

A 10-point scale was chosen to assess the quality of alternative theories where:

1 – very low quality and 10 is very high while 0 renders the theory irrelevant.

Additionally, we employ 5 parameters for theories assessment where each is supplied with set of weights. Parameters are:

Table 1. Weights of parameters' importance

weight	value
1	Low importance
2	Moderate importance
3	Hight importance

Table 2. Parameters with added weights.

Parameter	weight
Relevance	1

Applicable to the problem	2
Considers disability / discrimination	3
Considers event driven changes	3
Considers labor market	3

Table 3. Comparative table of alternative theories.

	Relevance (1)	<i>Applicable to the problem (2)</i>	<i>Considers disability discrimination (3)</i>	<i>Considers / event driven changes (3)</i>	<i>Considers labor market (3)</i>	SUM
<i>Human capital theory (HCT)</i>	8	18	12	15	27	80
<i>Demand for health capital</i>	7	14	15	18	18	72
<i>Institutional Ableism</i>	9	16	18	24	21	88
<i>Labor Market Segmentation Theory</i>	10	14	30	27	24	105
<i>Capability Approach</i>	10	12	27	24	21	94

As a result – the most scored framework of all five presented is **Labor Market Segmentation Theory** (Table 3). This approach incorporates socioal model of disability well and adequate by all presented metrics. It addressess discrimination of disabled people by both structural and institutional dimentions. It considers shock events as drivers of structural changes in secondary sector of employment. We will

be using the theory proposed by Peter Doeringer and Michael Piore whose studies have demonstrated their applicability and relevance in this context.

1.3 Application of labor market segmentation theory to analyze impact of pandemic restrictions on disabled person income

Labor Market Segmentation Theory (LMST) provides a powerful analytical framework for understanding why disabled workers experience sharper relative wage declines during economic crises compared to their non-disabled counterparts. Central to LMST is the distinction between the primary and secondary labor markets. The primary sector is characterized by stable employment, high wages, job security, fringe benefits, and internal labor markets with promotion ladders. In contrast, the secondary sector offers low wages, precarious contracts, limited benefits, high turnover, and little to no job protection. Disabled workers, due to persistent structural discrimination, lower average educational attainment resulting from earlier exclusion, and employer perceptions of lower productivity or higher accommodation costs, are disproportionately sorted into secondary sector jobs. This overrepresentation means that their employment conditions are intrinsically tied to the volatility of the secondary market.

In this context, secondary sector is not merely a repository of low-paid work but a shock-absorbing buffer for the wider economy. Regularly, wage differentials between primary and secondary sectors remain relatively stable, as secondary sector wages are often determined by minimum wage floors, supply and demand for low-skill labor, and institutional factors. However, in a crisis such as a pandemic-induced lockdown firms in the secondary sector face immediate demand contractions, credit constraints, and the need for rapid cost reduction. Unlike primary sector employers, who may resort to internal adjustments such as wage freezes, reduced hours with benefits, or temporary layoffs with rehiring commitments, secondary sector

employers have no such buffers. Their response is swift and brutal: wage cuts, contract non-renewals, and outright dismissals. Because disabled workers are overrepresented in this sector, they bear a disproportionate share of these adjustments, leading to a measurable fall in their average wages relative to non-disabled workers, who remain more sheltered in the primary sector.

Crucially, LMST posits that the secondary sector operates under fundamentally different institutional logics during crises compared to ordinary times. In ordinary times, secondary sector wages may be “sticky” at the bottom due to statutory minimum wages or social norms, but these minima do not prevent cuts in hours, overtime, or piece rates. More importantly, during a crisis, the equilibrium of the secondary market shifts: the labor supply expands as displaced primary sector workers temporarily accept secondary jobs, while labor demand contracts. Since secondary sector jobs already feature weak bargaining power and high substitutability, employers can unilaterally reduce real wages without fear of organized resistance. Disabled workers, facing those barriers have to rely on social system resources more heavily to compensate lost earnings.

During first wave of restrictions in March 2020 through economic disruption, the divergent behavior of the primary and secondary labor markets became starkly visible, directly informing the wage decline of disabled workers relative to non-disabled workers. In the primary most non-disabled workers retained their employment, often transitioning to remote work with preserved wages, supported by government-mandated paid leave and enterprise-level collective agreements. By contrast, the secondary sector underwent a sudden and deep contraction; disabled workers, who were overrepresented in these precarious roles due to prior hiring discrimination and limited access to primary sector employment, faced immediate consequences such as unpaid leave. Consequently, disabled workers experienced wage fall much harsher compared to non-disabled workers.

In order to measure first wave of pandemic restrictions we will employ difference in differences method on average earnings in order to assess wage decline.

2. Empirical study of the impact of COVID-19 pandemic on employment outcomes of workers with disabilities.

2.1 Analysis of empirical approaches to assess outcome variables and socio-demographic factors of disability as social status.

This section will review empirical research on the employment outcomes during pandemic for people with disabilities. Additionally we will research socio-demographic factors that influence disability.

Jones in 2022 explores the differential labour market impact of the first wave of the COVID-19 pandemic on disabled people in the UK through the lens of a two way fixed effects difference in differences, employing data from the Quarterly Labour Force Survey (QLFS) for 2019 and 2020³². The pre-pandemic analysis utilises 2019 data to estimate disability gaps in COVID-19-related economic and health risks, modelling each risk factor for individual i using OLS linear probability models as:

$$R_i = \alpha + \mu D_i + \gamma P_i + \varepsilon_i \quad 2.1$$

where

- R_i is a risk factor.
- D_i is a binary indicator of disability.
- P_i includes personal characteristics such as gender, age, marital status, presence of dependent children, highest qualification, ethnicity, and region.

The analysis finds that disabled workers faced higher risks, including a greater probability of being employed in ‘shutdown’ industries and in occupations with greater proximity to others and exposure to disease, even after accounting for these covariates.

³² Jones, M. (2022). COVID-19 and the labour market outcomes of disabled people in the UK. *Social Science & Medicine*, 292, Article 114637. <https://doi.org/10.1016/j.socscimed.2021.114637>

The main analysis then estimates the change in labour market outcomes between the pre-pandemic period (2019) and the post-lockdown period (April–December 2020) using a difference-in-differences specification:

$$L_{it} = \alpha + \mu D_{it} + \theta Post_t + \beta D_{it} Post_t + \gamma P_{it} + \delta W_{it} + \varepsilon_{it} \quad 2.2$$

where:

- L_{it} is labour market outcomes for individual i in year t .
- D_{it} is a binary indicator of disability.
- W_{it} work-related characteristics: including part-time employment; self-employment (where relevant); months tenure with current employer (and tenure squared) and sector.
- P_{it} includes personal characteristics such as gender, age, marital status, presence of dependent children, highest qualification, ethnicity, and region.

The dependent variables include employment status, hourly pay, being temporarily away from work (a proxy for furlough), and working reduced hours. Jones finds that established measures of inequality, including the disability employment gap (DEG) and disability pay gap (DPG), showed limited change during 2020. However, the increase in the probability of being temporarily away from work was approximately 40% larger for disabled workers even after controlling for work-related characteristics such as part-time status, self-employment, tenure, sector, and major occupation and industry groups.

To approach a causal interpretation, Jones tests the parallel trends assumption by extending the pre-COVID-19 period to 2013 and including a time trend interacted with disability, finding that the results are robust. She concludes that while the disability employment and pay gaps remained stable during the first wave, the disproportionate use of the furlough scheme among disabled workers—indicated by the higher probability of temporary absence—represents a hidden form of labour

market disadvantage that could have longer-term consequences for future disability-related inequality.

Ne'eman and Maestas in 2022 explore the differential employment trajectories of people with disabilities (PWD) through the lens of a pre–post pandemic comparative and occupational-category analysis³³. They made use of monthly data from the nationally representative Current Population Survey (CPS) spanning July 2008 to June 2022, with primary analyses beginning in July 2012 and focusing on the COVID-19 public health emergency from Q1 2020 to Q2 2022. The sample is restricted to working-age adults (18–64 years), with disability status defined using the standard CPS six-question sequence (covering hearing, vision, cognitive, ambulatory, self-care, and independent living difficulties). Their dependent variable is the employment-to-population ratio, calculated as the number of employed persons “at work” in a disability status group in a given month or quarter, divided by the total number of persons in that disability status group within the same period. To understand heterogeneous effects, they disaggregated occupations into three binary categories: teleworkable vs. non-teleworkable (based on the Dingel–Neiman classification), essential vs. non-essential (based on Department of Homeland Security guidelines), and frontline (essential and not teleworkable) vs. non-frontline (any occupation not meeting both essential and non-teleworkable criteria).

The econometric strategy employs linear probability models (LPMs) to estimate and compare percentage changes in employment-to-population ratios. The primary specification regresses the monthly employment indicator for individuals in a given disability–occupation cell on a dummy for disability status, quarter fixed effects, and their interaction, while also including a rich set of demographic control variables aggregated to the monthly level: age group (18–34, 35–49, 50–64), sex,

³³ Ne'eman, A., & Maestas, N. (2023). How has COVID-19 impacted disability employment?. *Disability and health journal*, 16(2), 101429. <https://doi.org/10.1016/j.dhjo.2022.101429>

race/ethnicity (white, Black, Hispanic, other), and educational attainment (Bachelor's degree or less). To avoid anomalous baseline effects in Q1 2020 (when a sharp, pre-pandemic spike in employment among PWD in non-essential occupations was observed), the authors set Q1 2019 as the reference quarter, with Q1 2020 included only as a placebo interaction term. Their general equation can be written as:

$$\ln(EPOP_{td}) = \alpha + \pi D + \sum_q \gamma_q Q_t + \sum_q \beta_q (Q_t \times D) + \delta' X_{td} + \varepsilon_{td} \quad 2.3$$

where:

- $EPOP_{td}$ is the employment-to-population ratio in month t for disability status group d.
- Q_t is a series of indicator variables for quarters across the study period (omitting the reference quarter Q1 2019).
- D is an indicator variable taking the value 1 when the monthly employment-to-population ratio observation is for disabled workers and 0 when it is for non-disabled workers.
- X_{td} is the set of group-specific, time-varying demographic covariates such as: age group (18–34, 35–49, and 50–64), sex, race/ethnicity (white, Black, Hispanic, or other), and educational attainment (no Bachelor's degree or Bachelor's degree).

The main findings diverge sharply from earlier recessionary patterns. As the COVID-19 recession began in Q2 2020, people with disabilities experienced employment losses that were proportionately similar to those of people without disabilities, in contrast to the greater losses observed during the Great Recession. However, during the subsequent economic recovery, the employment rate of people with disabilities grew much more quickly: by Q4 2021 through Q2 2022, the

employment-to-population ratio for PWD exceeded its pre-pandemic level by nearly 3.6 percentage points, whereas the rate for persons without disabilities remained approximately 0.5 percentage points below its Q1 2019 baseline. Notably, these employment gains were concentrated precisely in teleworkable, essential, and non-frontline occupations, with the coefficient on the interaction term Disabled $\times$ Q2 2022 reaching 0.186 ($p < 0.01$) for teleworkable jobs and 0.169 ($p < 0.01$) for non-frontline jobs. Finally, the authors confirm that this outperformance is driven by increased labor force participation among people with disabilities rather than by a compositional change in the unemployment rate, suggesting that the pandemic-induced shift to remote work may have opened new, durable labour market opportunities for this historically disadvantaged group.

Maroto and Pettinicchio explore how rates of job displacement varied for individuals with disabilities through the lens of a large-scale, longitudinal quantitative analysis spanning two major economic crises: the 2008 Great Recession and the first wave of the COVID-19 pandemic³⁴. Their study uses data drawn from six waves of the U.S. Current Population Survey's Displaced Worker Supplement (CPS DWS), a nationally representative survey that collects detailed information on workers who have lost or left jobs due to plant or company closures, slack work, or the elimination of positions. The CPS DWS also includes a standard set of six disability questions covering difficulties in hearing, vision, cognition, ambulation, self-care, and independent living. The final analytical sample consists of $N = 344,729$ respondents. To model the probability of experiencing involuntary job displacement, the authors estimate a series of logistic regression models. The general form of the logistic model can be expressed as:

³⁴ Maroto, M., & Pettinicchio, D. (2024). From recession to pandemic: Displacement among workers with disabilities from 2007 through 2021. *Journal of Vocational Rehabilitation*, 61(1), 25–37. <https://doi.org/10.3233/JVR-230064>

$$\ln\left(\frac{p_i}{1-p_i}\right) = \beta_0 + \beta_1(disability\ type_i) + \beta_2Period_i + \beta_3(disability\ type_i \times Period_i) + \gamma X_i + \varepsilon_i \quad 2.4$$

where:

- p_i is the probability of being displaced for individual i .
- *Disability type* is a set of dummy variables capturing both the presence and type of disability (e.g., cognitive, ambulatory, independent living).
- *Period* includes dummy variables for the Great Recession (2007–2009), the pandemic period (2020–2021), and the recovery years in between.
- X_i is a vector of individual-level controls (age, gender, race, educational attainment, industry, and occupation).

The interaction term β_3 is the key coefficient, measuring the differential effect of each economic crisis on the probability of job displacement for people with disabilities relative to their non-disabled peers.

The main findings reveal a persistent and crisis-magnified pattern of inequality. People with disabilities were approximately twice as likely as those without disabilities to experience job displacement over the entire study period. The disparity was especially pronounced during times of economic turmoil. Although the displacement gap by disability status had been declining—from a high of 6.5 percentage points during the Great Recession to a lower point in the intervening years—the COVID-19 pandemic abruptly increased the gap to 5.8 percentage points. This finding shows that involuntary job loss among people with disabilities is exacerbated by exogenous shocks, a pattern that holds across multiple disability types. The authors conclude by extending prior research on disability and labour market precarity, situating the COVID-19 pandemic as a critical case where pre-existing vulnerabilities are amplified, and they call for policy interventions that

address both the immediate and the structural drivers of disability-based displacement.

In terms of disability measurement, the study disaggregates the general “any disability” indicator into specific types: hearing impairment, vision impairment, cognitive difficulty, ambulatory (mobility) difficulty, self-care difficulty, and independent living difficulty. This disaggregation yields the finding that disparities were largest for individuals with cognitive disabilities, for whom the probability of displacement during the pandemic was particularly elevated. Women and members of racialised minority groups with disabilities faced intersectional disadvantage, with displacement rates exceeding the sum of their constituent parts.

From a policy perspective, the authors note that the CPS DWS does not capture discrimination directly. However, the presence of large and persistent gaps even after controlling for industry, occupation, and worker characteristics is consistent with ongoing structural discrimination—formalised as hiring bias, unequal access to telework accommodations, and lower seniority protections among workers with disabilities—that becomes more consequential when the labour market tightens.

Bondoux in 2024 explores the complex, bidirectional relationships between disability and labour market outcomes across Europe through the lens of causal econometric methods applied to longitudinal panel data. Her thesis utilises data from the Survey of Health, Ageing and Retirement in Europe (SHARE), which provides a large sample of individuals followed over time across 28 European countries³⁵. The study employs variables including personal income (defined as the sum of wages and compensation incomes), disability onset status, partner's disability status, labour supply (measured as hours worked or employment status), unemployment

³⁵ Bondoux, Justine, (2024), Relationships between disability and labour market outcomes in Europe

spells, and national social security system characteristics such as integration measures and compensation income generosity. She estimates the causal impact of disability onset on personal income, she implements a matching procedure before disability onset followed by a weighted difference-in-differences (DiD) estimator. Propensity score matching covariates included variables such as: age, gender, education level, marital status, having children and occupational sector. Logit model was used to estimate the individual probability of being treated on a set of observed individual characteristics X : age, gender, education level, being in a couple, having at least one child, being in the private or public sector, and being self-employed. Weighted difference in difference model included specification:

$$\ln(\text{income} + 1)_{it} = \beta_0 + \beta_1 \text{treated}_i + \beta_2 \text{after}_t + \beta_3 (\text{treated} \times \text{after})_{it} + \beta_4 \text{job situation}_{it} + \delta_i + u_{it} \quad 2.5$$

where:

- income_{it} is either the personal income, the wages or the compensation incomes of individual i in time t
- treated_i takes the value 1 if individual i is in the treatment group.
- $\text{job situation}_{it}$ is a four-category variable consisting of employed or self-employed, retired, unemployed, and other situation (which groups together permanently sick or disabled individuals, homemakers, etc.).
- δ_i is an unobserved time-invariant individual effect.

This approach combines PSM's ability to reduce selection on observables with DiD's ability to eliminate time-invariant unobserved heterogeneity common to both groups. The identifying assumption, therefore, is that conditional on the propensity score and in the absence of disability onset, the treatment and matched control groups would have followed parallel trends in income. Bondoux validates this by

testing for pre-treatment parallel trends and by examining that the matching procedure successfully balances covariates (standardised differences are reduced below 10% after matching). The finding of a significant negative effect of disability onset on personal income (robust to placebo tests, alternative matching algorithms, and different bandwidth choices) is thus interpreted as causal evidence of the labour market penalty associated with becoming disabled in Europe, mitigated only in countries with generous integration measures and high compensation incomes.

Schur, van der Meulen Rodgers, and Kruse explore the disparate effects of the COVID-19 pandemic on workers with disabilities through the lens of an intersectional econometric decomposition approach³⁶. Using monthly data from the U.S. Current Population Survey (CPS), the authors restrict the sample to working-age adults (18–64 years) with strong labour market attachment (currently employed or employed within the prior 12 months). The dependent variable is a binary employment indicator (employed vs. not employed) measured across three key pandemic periods: January–March 2020 (pre-lockdown), April–December 2020 (post-lockdown), and separately January–April 2020 (to capture the initial drop). Disability status is identified using the standard six-question CPS sequence covering hearing, vision, cognitive, ambulatory, self-care, and independent living difficulties. The core independent variables include categorical measures of gender (male/female), race/ethnicity (White non-Hispanic, Black, Hispanic, Asian, other), and the three-way interaction among disability status, gender, and race/ethnicity. Additional demographic controls include educational attainment (less than high school, high school, some college, Bachelor’s degree, advanced degree), marital status, and age. Occupational and industry controls are coded according to standard CPS classifications, with particular attention to sectors that were hardest hit by the

³⁶ Schur, Lisa & Rodgers, Yana & Kruse, Douglas. (2021). COVID-19 and Employment Losses for Workers with Disabilities: An Intersectional Approach. SSRN Electronic Journal. 10.2139/ssrn.3788319.

initial pandemic shutdowns. All estimates are seasonally adjusted by the authors and reweighted to account for disability-specific seasonal patterns, and the analysis employs CPS sample weights throughout.

The main findings substantiate the initial intersectional hypothesis. The raw disability employment gap (persons with disabilities minus persons without disabilities) widened from 5.9 percentage points in the pre-lockdown period (January–March 2020) to 7.6 percentage points in the post-lockdown period (April–December 2020), a statistically significant increase of 1.7 points. Decomposition results show that 40% of this increase is explained by observable characteristics, with occupational distribution accounting for the largest share, followed by education. Crucially, even after accounting for differences in occupation, industry, education, and demographics, there remains a substantial unexplained component of the employment gap that also rose during the pandemic, which the authors interpret as a potential signal of growing discrimination against people with disabilities by employers. When examining three-way interactions, the logit results identify that White and Black women with disabilities experienced the most severe and persistent employment losses during 2020 relative to White men without disabilities, and Hispanic men with disabilities also bore a heavy burden, though the latter finding is less robust due to small sample sizes. The study's novel contribution lies in its disaggregation of the pandemic's shock across multiple axes of disadvantage, demonstrating that the labour market penalty for disability is not uniform but is substantially exacerbated for those who also experience gender and racial marginalisation.

Table 4. Conclusions obtained as a result of a literary review of empirical works.

Paper	Brief conclusions
<i>Jones (2022)</i>	Linear probability difference in differences used. Disabled workers faced higher risks, including a greater probability of being employed in ‘shutdown’ industries.
<i>Ne’eman and Maestas (2022)</i>	Dynamic linear probability difference in differences used. During pandemic people with disabilities experienced employment losses that were proportionately similar to those of people without disabilities. During the subsequent economic recovery, the employment rate of people with disabilities grew much more quickly.
<i>Maroto and Pettinicchio (2024)</i>	Logistic regression difference in differences used. People with disabilities were approximately twice as likely as those without disabilities to experience job displacement over the entire study period
<i>Bondoux (2024)</i>	Weighted difference in differences used with propensity score matching. Significant negative effect of disability onset on personal income (robust to placebo tests, alternative matching algorithms, and different bandwidth choices) is thus interpreted as causal evidence of the labour market penalty associated with becoming disabled in Europe. PSM includes socio-demographic factors as well as work related.

<i>Schur, van der Meulen Rodgers, and Kruse (2021)</i>	Intersectional econometric decomposition approach is used. The raw disability employment gap (persons with disabilities minus persons without disabilities) widened from 5.9 percentage points in the pre-lockdown period (January–March 2020) to 7.6 percentage points in the post-lockdown period (April–December 2020), a statistically significant increase of 1.7 points.
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Overall most empirical studies rely either on employment probabilities or income regressions. Disability definitions differs substantially and mostly rely on self-reported metrics. Crucial role of market sectors is highlighted as many of disabled people occupied mainly ‘shutdown’ industries with higher risks of unemployment during the pandemic. Propensity scores methods rely on socio-demographic factors such as gender and age coupled with work environment variables to indicate industry or conditions. Review of empirical studies provides a deeper understanding of application of various methods which are summarized in the table 4

2.2 Data overview and variable selection

We use Russian Longitudinal Monitoring Survey panel data. RLMS aim to monitor the effects of Russian reforms on the health and economic welfare of individuals in the Russian Federation. We use individual adult panel with post-stratification weights. With period of interest 2019 to 2020 we also included years from 2016 to 2022 to address parallel trends assumption.

Following sample selection procedures were performed:

- *Interviews from 4th annual quaters were selected:* this condition is based both on wave collection periods (usually September to January) and initial pandemic lockdown at the end of 2020.
- *Only adults were selected.*
- *Individuals with extreme earnings were excluded:* using calculated combined job income (formula) we applied threshold at 95th percentile and excluded selected outliers.
- *Only people present in 2019 and 2020 and employed at least in one of the periods were selected:* that condition ensures that only employed at the period people were selected reflecting both possible job loss and gain and allows panel method application due to constant presence of stable units.
- *Spillovers were excluded:* people who acquired or lost disability status during period of interest were excluded due to violation of SUTVA requirements: control units may be affected by the treatment.

Selected panel data included following variables:

- *Wage_j1* – after tax wages at the first job in last 30 days. This is a self reported variable in rubles that reflects wages earned at the first or only job of individual.
- *Wage_j2* – after tax wages at the second job in last 30 days. This is a self reported variable in rubles that reflects wages earned at the second job of individual. Situations possible were first job earnings may be zero with real earnings from additional job.
- *Income_iea* – self reported additional paid work income in last 30 days. This variable reflects all income gained aside listed previously wages. It reflects all additional unofficial incomes received by individual for completed work (e. g. services of nurse, food delivery or repairing).

- *Age* – self reported number of full years by individual.
- *Education_details* – self reported level of education received by respondent over 14 years. It includes levels such as: years of school, specialized school, university etc. We choose to include university level only because of higher influence on outcome. Final variable is a dummy.
- *Marital_status* – self reported marital status with levels such as “married”, “never married” etc. The variable was reduced to 1: “married”, “living together” and 0: “never married”, “divorced”, “widow”. Final variable is a dummy.
- *Gender* – self reported gender. Transformed into a dummy with 1 = is_female.
- *Harmful_work* – reported dangerous, harmful or unhealthy conditions of workplace at job 1. Is a dummy variable.
- *Current_work_status* – self reported work status of the individual. It contains variables such as “you are currently working”, “you are not working”, “you are on paid leave” etc. The variable was transformed into dummy variable 0 – not working and 1 – currently working.
- *Has_disability* – reported assigned disability class or absence of it. We consider this variable *objective* because the unlike self reported health variables about individual’s faculties this variable was assigned by external organization and self report bias may be hidden only in reporting variable itself. The variable was transformed into a dummy with 1 standing for “has disability” and 0 otherwise. Cases of changing disability status were excluded at the period of interest.

Table 5. Continuous variable descriptive weighted statistics.

<i>Variable</i>	<i>Weighted Mean</i>	<i>Weighted STD</i>
<i>After tax wages at first job in last 30 days</i>	26659.08	19298.23
<i>After tax wages at second job in last 30 days</i>	348.21	2779.33
<i>Additional paid work income in last 30 days</i>	460.39	3520.9
<i>Age</i>	42.62	12.17

Table 6. Binary variable descriptive weighted statistics.

<i>Variable</i>	<i>weighted frequencies</i>	
	Var = 0	Var = 1
<i>Education details: University</i>	5899.47	3696.51
<i>Marital status: Married</i>	2614.79	6981.19
<i>Gender: Is female</i>	4766.761	4829.224
<i>Harmful work: Yes</i>	8736.53	859.45
<i>Current Work Status: Currently works</i>	857.26	8738.7
<i>Has disability: Yes</i>	9802.07	222.89

Additional metric *wages* is calculated based on 3 main employment earnings (formula):

$$wages = wage_{j1} + wage_{j2} + income_{iea} \quad (2.6)$$

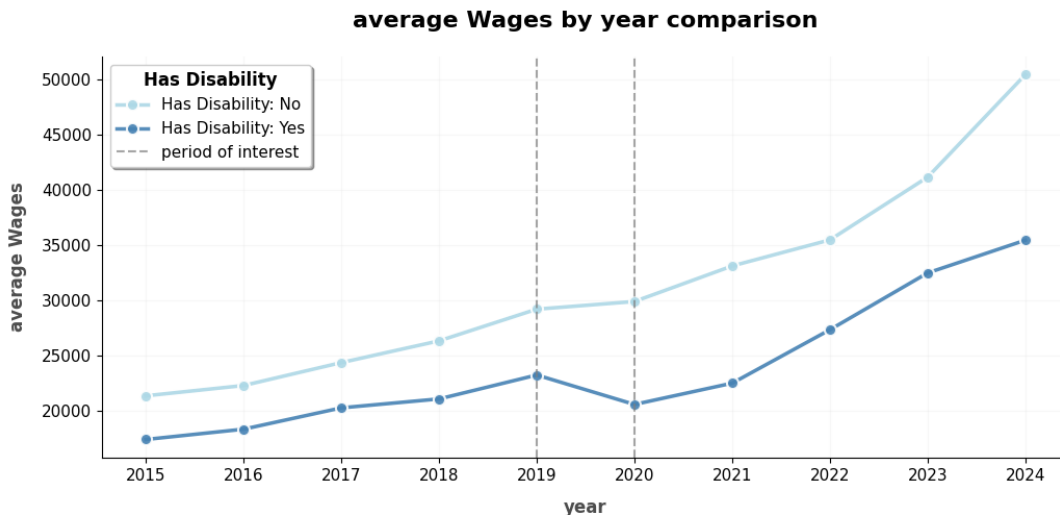
where:

- $wage_{j1}$ – reported wages from first job,
- $wage_{j2}$ – reported wages from second job,

- $income_{iea}$ – reported additional earnings gained by paid labor.

This metric combines total employment earnings reflecting both official employment and unofficial sources of income. Additionally, this approach resolves situations with unpaid first jobs where individuals report zero received earnings

Picture 1 shows timelapse of average wages for disabled and non-disabled groups. At the period of interest (2020) we observe a dip caused by pandemic. While it affected both groups altogether disabled people has suffered greater fall compared to control units.



Picture 1. Timelapse comparison of average wages between control units (non-disabled people) to treatment units (disabled people)

2.3 Developing TWFE DiD model with propensity matching approach.

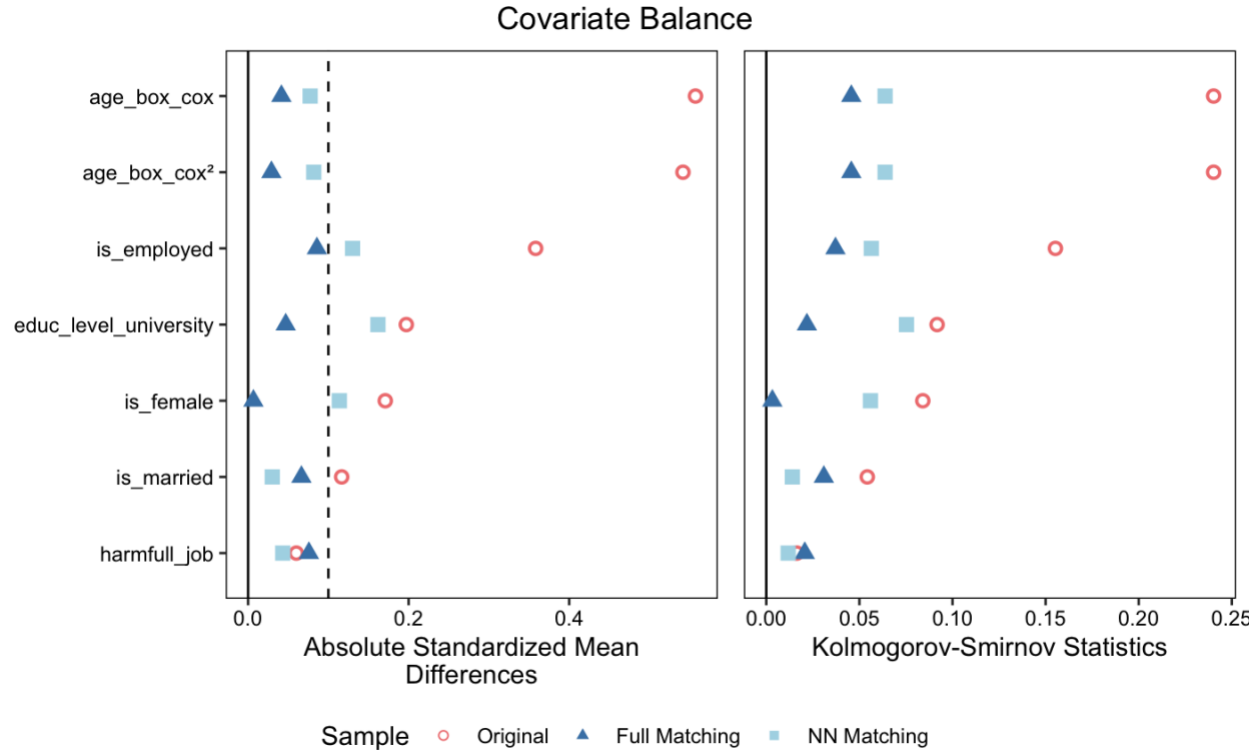
As presented in the descriptive statistics (Table ##), the *has_disability* variable exhibits markedly disproportionate frequencies across comparison groups. This imbalance, coupled with the absence of random assignment – which is infeasible in disability research due to the inherent nature of disability status—poses

a fundamental threat to valid causal inference in observational studies . Under non-randomized designs, treated and untreated units cannot be presumed comparable on pretreatment covariates, yet the validity of causal claims critically depends on such equivalence. To address these methodological challenges, we employ Propensity Score Matching. The propensity score, proposed by Rosenbaum and Rubin's seminal work, demonstrates that conditioning solely on this scalar propensity score suffices to eliminate bias attributable to all observed covariates, thereby approximating the conditions of a randomized experiment in observational studies³⁷.

We employed variables: age, education level, gender, harmful job and employment to match units properly. To address non-normality of age distribution we performed Box-Cox transformation on the age variable with resulting $\lambda = 0.303$. Variable *Age_Box_Cox* was calculated according to formula 2.7

$$Age_{box-cox} = \frac{Age^{(\lambda-1)}}{\lambda} \text{ where } \lambda = 0.303 \quad (2.7)$$

³⁷ Rosenbaum, P. R., & Rubin, D. B. (1983). The central role of the propensity score in observational studies for causal effects. *Biometrika*, 70(1), 41–55.



Picture 2. Covariate balance comparison between multiple psm approaches (distances reflected with ASMD and Kolmogorov-Smirnov metrics).

We performed several types of propensity score matching in order to identify the most appropriate for the case (Picture 2):

- nearest neighbor 1:1 matching without replacement with a propensity score estimated using logistic regression of the treatment on the covariates
- logit optimal full matching estimated with logistic regression.

Nearest-neighbor matching yielded poor balance as indicated in Picture 2. On the contrary logit model has yielded adequate balance. The propensity score was estimated using a logit regression of the treatment on the covariates, which yielded better balance than did NN approach. After matching, all standardized mean differences for the covariates were below 0.1 and all standardized mean differences for squares and two-way interactions between covariates were below .15, indicating

adequate balance. Full matching uses all treated and all control units, so no units were discarded by the matching. Panel weights were applied to generate PSM weights. Tables ## and ## summarize balancing results of full logit model approach. Low standardized mean differences and eCDF statistics indicate adequate balancing performed by PSM approach. As shown in table ## total of 221 of treated individuals were matched to 9309 of control individuals.

Final PSM specification as following form (formula ##):

$$\log\left(\frac{P(disability = 1)}{1 - P(disability = 1)}\right) = \beta_0 + \beta_1 \cdot age_{box\ cox} + \beta_3 \cdot education: university + \beta_4 \cdot female + \beta_5 \cdot harmful\ job + \beta_6 \cdot employed \quad (2.8)$$

where:

- *disability*: is outcome variable *has_disability* considered as a probability of the treatment. Social model of disability was elected in order to evaluate treatment probability as environmental and social factors were considered to be regressors.
- *Age_box_cox*: Box-Cox transformed *Age* variable using $\lambda = 0.303$.
- *education level: university*: achieved education level: university.
- *female*: female gender of the respondent.
- *harmful job*: harmful environment exposure at working place.
- *employed*: employment status.

Table 7. Propensity score balancing results summary. Low eCDF and SMD indicate adequate balancing strategies

Variable	Means treated	Means control	Std, mean diff.	Var. Ratio	Ecdf mean	Ecdf max	Std. Pair dist.
Distance	0,035	0,033	0,072	1,037	0,019	0,070	0,015
Age_{box-cox}	7,358	7,323	0,042	0,817	0,015	0,046	0,861
Education:university	0,317	0,339	-0,047		0,022	0,022	0,921
Female	0,410	0,407	0,007		0,003	0,003	0,756
Married	0,683	0,652	0,066		0,031	0,031	0,908
Harmfull job	0,081	0,102	-0,076		0,021	0,021	0,523
Employed	0,749	0,786	-0,086		0,037	0,037	0,292

Table 8. Propensity score matching results summary. Total 9530 were matched.

	control	treated
all (ESS)	9177,95	215,6
all	14595	325
matched (ESS)	349,21	215,6
matched	9309	221
unmatched	5286	104
discarded	0	0

Two way fixed effects OLS was performed with clustered standard errors at individual level. Clustering of the covariance was performed due to presence of significant serial correlation with coefficient on lagged differences equals to -0,447 and z-statistic equals to -9,342 at 99% level of significance. Wooldridge's test was used for serial correlation in panel data (table 10). Serial correlation presence violates BLUE assumptions about ordinary least squares and thus must be eliminated by means of covariance type adjusting at individual level.

Following two way fixed effects (TWFE) specification is used:

$$wages_{it} = \gamma_i + \lambda_t + \delta \cdot disability_{it} \times period_{it} + \varepsilon_{it} \quad (2.9)$$

where:

- $wages_{it}$ – computed aggregated job income for individual i in period t .
- $disability_{it} \times period_{it}$ – treatment indicator of being disabled at period = 1 (the beginning of the pandemic).
- δ – effect of being disabled at period = 1 (the beginning of the pandemic).
- γ_i – unit specific fixed effects of being disabled.
- λ_t – time-specific fixed effects of the pandemic.
- ε_{it} – error term where $E[\varepsilon_{it}] = 0$.

As indicated in table ##, the average marginal effect of having a disability on aggregated wages during the pandemic period is a statistically significant decline of -5,678.95 in monthly earnings compared to their pre-pandemic baseline relative to non-disabled workers (SE = 2,350.19, p = 0.016). The 95% confidence interval ranges from -10,287.58 to -1,070.32. This corresponds to approximately a 19% reduction relative to the pre-pandemic mean for this group.

Table 9. TWFE summary statistics for total market.

variable	parameter	std. Err.	t-stat	p-value	upper CI	lower CI
has_disability × period	-5678,95*	2350,19	-2,41638	0,01575	- 10287,58	-1070,319

Num.Obs.	4772
R2	0,659
R2 Adj.	0,552
R2 Within	0,007
R2 Within Adj.	0,007
Fixed-effects	2386
AIC	222986,5
BIC	240224,0
RMSE	13835,63
Time Effects	True
Entity Effects	True
Covariance type	Clustered (ind)

. p < 0.1, * p < 0.05, ** p < 0.01, *** p < 0.001

Overall R^2 0.659 indicates that the fixed effects alone absorb substantial cross-sectional variation in earnings.

Table 10. Wooldridge's test for serial correlation in panel data results.

variable	parameter	std. Err.	z	p-value	upper CI	lower CI
Intercept	3335,742***	281,878	11,834	0,000	2783,272	3888,213
resid_diff_lag	-0,447***	0,048	-9,342	0,000	-0,541	-0,353

Num.Obs.	5188
R2	0,162
R2 Adj.	0,161
AIC	117201,014
BIC	117214.1219
Covariance type	HC1
. p < 0.1, * p < 0.05, ** p < 0.01, *** p < 0.001	

To address parallel trends assumption a dynamic TWFE model was computed with following specification:

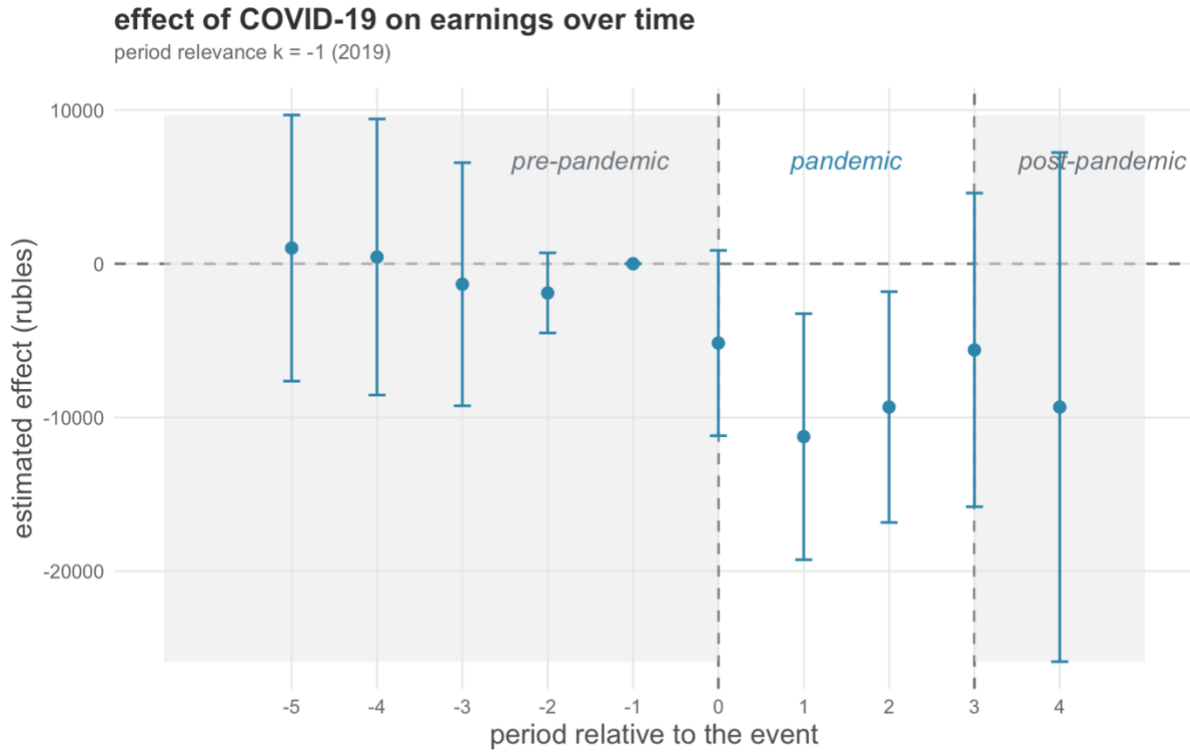
$$wages_{it} = \gamma_i + \lambda_t + \sum_{k=-l}^m \delta_k \cdot disability_{ki} \times period_{ki} + \beta \cdot X'_{it} + \varepsilon_{it} \quad (2.10)$$

where:

- $wages_{it}$ – computed aggregated job income for individual i in period t .
- $disability_{ik} \times period_{ik}$ – treatment indicator of being disabled at period = k (the beginning of the pandemic).
- δ_k – effect of being disabled at period = k .
- l – the number of pre-treatment periods.
- m – the number of post-treatment periods.
- γ_i – unit specific fixed effects of being disabled.
- λ_t – time-specific fixed effects of the pandemic.
- ε_{it} – error term where $E[\varepsilon_{it}] = 0$.

Table 11 demonstrates no significant effects prior to the pandemic. Additionally, parameters comparatively close to zero indicating prominent parallel trends picture

##. Joined F-test for pretreatment period shows no significant statistic (table ##):
 $F = 2.02319$. p-value = 0.098783 with 1,603 DoF.



Picture 3. Coefficient plot of dynamic TWFE regression with highlighted pandemic period.

Table 11. Results of dynamic TWFE.

variable	parameter	std. Err.	t-stat	p-value	upper CI	lower CI
period_relevance = -5 × has_disability × period	1020,673	947,342	1,0774	0,2814	-4119,58	14160,93
period_relevance = -4 × has_disability × period	437,918	4577,347	0,0956	0,9237	-8540,3	9416,131
period_relevance = -3 × has_disability × period	-1332,546	4031,562	-0,3305	0,7410	-9240,23	6575,142
period_relevance = -2 × has_disability × period	-1899,068*	-839,064	-2,2633	0,0237	-11011,3	-786,807

period_relevance = 0 × has_disability × period	-5158,921+	3073,242	-1,67866	0,0934	-11186,9	869,0732
period_relevance = 1 × has_disability × period	-11248,3**	4080,274	-2,75674	0,0059	-19251,5	-3245,03
period_relevance = 2 × has_disability × period	-9323,438*	3827,022	-2,43621	0,01495	-16829,9	-1816,94
period_relevance = 3 × has_disability × period	-5605,689	5201,050	-1,0778	0,2812	-15807,3	4595,885
period_relevance = 4 × has_disability × period	-9323,863	8146,257	-1,10437	0,26959	-25883,8	7236,054

Num.Obs.	9431
R2	0,646
R2 Adj.	0,572
R2 Within	0,002
R2 Within Adj.	0,001
Fixed-effects	1604
AIC	212586,5
BIC	224186,7
RMSE	15989,70
Time Effects	True
Entity Effects	True
Covariance type	Clustered (ind)

+ p < 0.1, * p < 0.05, ** p < 0.01, *** p < 0.001

Empirical study highlighted that disabled workers experienced a statistically significant decline in monthly employment earnings of -5,678.95 rubles (SE = 2,350.19, p = 0.016) during the pandemic period relative to their pre-pandemic baseline when compared to non-disabled workers. This decline represents

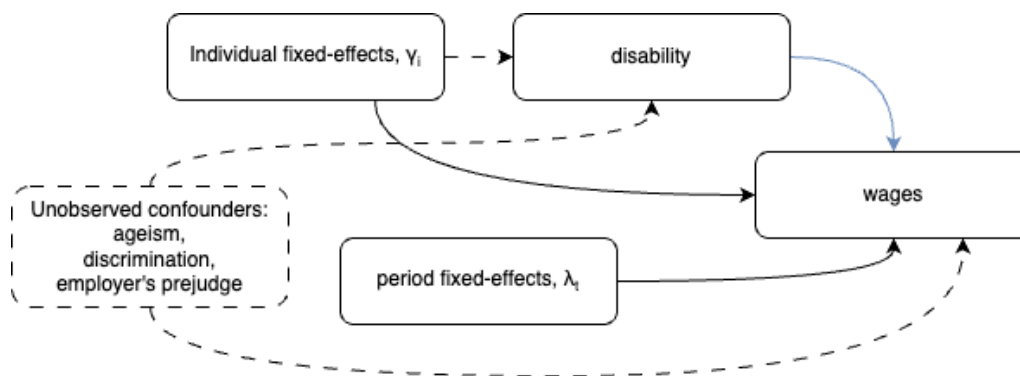
approximately a 19 percent reduction relative to the pre-pandemic mean for the treatment group. The 95 percent confidence interval, ranging from -10,287.58 to -1,070.32 rubles has reasonable precision around this estimate, though the magnitude of the effect should be interpreted with attention to its substantial lower bound.

The dynamic TWFE specification provides support for the parallel trends assumption. Pre-treatment coefficients for periods -5 through -2 are not systematically different from zero, with the joint F-test for all pretreatment periods failing to reject the null of no differential pre-trends ($F = 2.023$, $p = 0.099$). This lends credence to the interpretation of the post-treatment coefficient as a causal effect rather than an artifact of pre-existing divergence between groups. Notably, the negative effect becomes pronounced immediately at period zero (coinciding with the pandemic onset) and persists through periods one and two, with coefficients of -5,158.92 ($p = 0.093$), -11,248.30 ($p = 0.006$), and -9,323.44 ($p = 0.015$) respectively. By periods three and four, the estimates revert toward statistical insignificance, although wide confidence intervals caution against strong claims regarding effect dissipation.

The Propensity Score Matching procedure employing full logit matching successfully balanced covariates between treatment and control groups, with all standardized mean differences below 0.10 and all squared and interaction terms below 0.15 post-matching. This balance strengthens the comparability of the comparison groups and reduces concerns about confounding by observed characteristics such as age, education, gender, marital status, harmful work exposure, and employment status.

Several limitations warrant consideration. First, the self-reported nature of wage variables may introduce measurement error, though there is little reason to suspect differential reporting bias between disabled and non-disabled respondents. Second, the exclusion of individuals who acquired or lost disability status during the

period of interest, while necessary for SUTVA compliance, limits generalizability to the stable disability population. Third, the relatively small number of treated units (221 matched disabled individuals) relative to controls (9,309) may affect precision, though the cluster-robust standard errors partially address concerns about dependence. Fourth, the analysis captures short- to medium-term effects only; longer-term adjustments in labor market outcomes remain unobserved. Additionally there are possible cofounders presented in picture ## that may effect treatment directly such as: discrimination by age (ageism), employer’s prejudice which is present in all sectors of the employment market and other types of discrimination. Unfortunately those confounders remain unobserved.



Picture 4. Casual inference on outcome.

Future research should examine the mechanisms underlying the observed effect should investigate whether similar patterns hold in other national contexts with different disability support systems and pandemic policy responses. Additionally, as longer-term data become available, researchers should assess whether the earnings gap persists, narrows, or widens beyond the immediate crisis period.

In summary, our study provides robust evidence that disabled workers in Russia bore a statistically and economically meaningful earnings penalty during the initial years of the COVID-19 pandemic relative to their non-disabled counterparts. The findings underscore the importance of disaggregating crisis impacts by disability status and of designing targeted interventions to prevent the exacerbation of pre-existing labor market inequalities during systemic disruptions.

3. Recommendations to address similar shock events

The evidence presented in this study—specifically, a statistically significant and economically meaningful decline of approximately 19% in monthly earnings for disabled workers following the COVID-19 pandemic there are direct implications for labor market and social protection policy. In light of these findings, three interrelated sets of policy recommendations emerge, targeting crisis preparedness, employer accountability, and data infrastructure.

First, the disproportionate earnings shock experienced by disabled workers suggests that general economic stabilisation measures, while necessary, are insufficient to prevent the exacerbation of pre-existing inequalities during systemic crises. Policymakers should therefore embed disability-specific safeguards into emergency response frameworks. One concrete mechanism is the introduction of automatic earnings stabilisation transfers: a supplemental income benefit triggered when a verified macroeconomic shock (e.g., a pandemic, financial crisis, or natural disaster) coincides with a documented wage decline below a pre-crisis threshold for disabled workers. Such a scheme would operate parallel to general unemployment or short-time work benefits, addressing the specific vulnerability identified in this study. Furthermore, crisis employment retention programmes (such as wage subsidy schemes) should include mandatory quotas for disabled workers and provide employers with additional financial support for workplace accommodations during the crisis period.

Second, the absence of systematic employer-level data on disability employment and pay outcomes constitutes a critical barrier to evidence-based policy. This study employed propensity score matching precisely because observational data on disability status are neither randomised nor routinely disaggregated. To move beyond post-hoc statistical adjustments, governments should mandate annual disability employment and pay gap reporting for employers

above a certain size threshold (e.g., 250 employees). Such reporting would require firms to disclose hiring, retention, and promotion rates for disabled workers, as well as median wage differences between disabled and non-disabled employees in comparable roles. Inspired by the gender pay gap reporting frameworks adopted in several European countries, this mechanism creates transparency and incentivises employers to identify and remediate internal disparities. To ensure compliance, public procurement contracts could be conditioned on satisfactory disability inclusion metrics, leveraging the state's purchasing power as a driver of structural change.

Third, the methodological challenges encountered in this study—particularly the non-random assignment of disability status and the need to exclude individuals with changing disability status to satisfy SUTVA. National statistical offices and longitudinal survey programmes, including the Russian Longitudinal Monitoring Survey, should expand the collection of disability-related variables with greater granularity. Recommended enhancements include:

- (a) capturing both stable and newly acquired disability status over the panel wave;
- (b) recording the type and severity of disability using standardised international classifications (e.g., Washington Group Short Set);
- (c) collecting information on workplace accommodations received; and
- (d) tracking monthly earnings at a finer temporal resolution to allow event-study designs with more precise timing.

These improvements would enable future researchers to estimate causal effects with reduced bias and allow policymakers to target interventions to the most vulnerable subgroups.

Finally, while the present analysis focuses on the short- to medium-term effects of a single crisis, the findings suggest a more durable principle: disability-

inclusive policy cannot be an ex-post addendum but must be designed ex-ante into economic governance structures. To operationalise this principle, governments should establish standing inter-ministerial working groups on disability and economic shocks, tasked with pre-drafting disability annexes to all major crisis response plans. Such annexes would specify, for example, accessible application procedures for income support, reserved places in retraining programmes, and direct communication channels with disabled persons' organisations. By institutionalising this foresight, policymakers can mitigate the recurring pattern of disproportionate harm documented in this study and move toward a more resilient and equitable labour market.

Conclusion

This study set out to examine whether disabled workers in the Russian Federation experienced differential wage outcomes during the COVID-19 pandemic relative to their non-disabled counterparts, and to situate these findings within the framework of Labor Market Segmentation Theory. The empirical analysis, based on panel data from the Russian Longitudinal Monitoring Survey (2016–2022) and employing a Two-Way Fixed Effects Difference-in-Differences design with Propensity Score Matching, yields clear evidence that the pandemic onset disproportionately harmed the earnings of disabled workers.

The study consisted of several main stages. In the first stage, a theoretical framework was established, drawing on Labor Market Segmentation Theory to conceptualise how disability status may operate as a primary labour market segmentation mechanism, particularly during exogenous shocks. Key theoretical propositions – specifically, that disabled workers are overrepresented in secondary labor market segments characterised by lower job security, weaker bargaining power, and greater vulnerability to economic downturns – were operationalised into testable hypotheses. In the second stage, a comprehensive sample selection procedure was implemented using RLMS data, including the exclusion of individuals with changing disability status to satisfy SUTVA, outlier trimming at the 95th percentile, and the construction of an aggregated wage variable combining first job, second job, and additional paid work income. Propensity Score Matching (full logit optimal matching) was employed to balance treated and control units on observed covariates, successfully achieving standardised mean differences below 0.10. In the final stage, a TWFE DiD model with clustered standard errors was estimated, followed by a dynamic TWFE specification to test the parallel trends assumption and trace the evolution of treatment effects over time.

The results of the review of theoretical and empirical literature align with the core predictions of Labor Market Segmentation Theory. Scholars have long documented that disability status serves as a basis for labour market stratification, with disabled workers facing barriers to entry into primary segment jobs characterised by stable employment, career progression, and adequate earnings. The pandemic, as an exogenous labour demand shock, was expected to amplify these pre-existing disparities. The empirical analysis confirms this expectation: disabled workers experienced a statistically significant decline in monthly earnings of 5,678.95 rubles (approximately 19 percent relative to pre-pandemic mean) compared to non-disabled workers, with the effect persisting through the first two post-treatment periods. The dynamic model reveals no significant pre-treatment coefficients, supporting the parallel trends assumption and strengthening the causal interpretation of the finding.

Based on the factors described above, the study confirms the following hypotheses derived from Labor Market Segmentation Theory:

- H1. Disabled workers experience a larger decline in monthly earnings during the COVID-19 pandemic period compared to non-disabled workers.
- H2. The earnings penalty experienced by disabled workers persists beyond the immediate onset of the pandemic (2020) into subsequent years (2021–2022), indicating that the shock produces a medium-term divergence in earnings trajectories rather than a temporary dip.

The theoretical contribution of this study lies in extending Labor Market Segmentation Theory to the analysis of exogenous shocks. While the theory has traditionally focused on persistent structural barriers, this research demonstrates that segmentation mechanisms are amplified during crises, producing widening rather than stable earnings gaps. Methodologically, the study contributes by demonstrating

the feasibility and necessity of combining Propensity Score Matching with TWFE DiD to address non-random assignment of disability status in observational panel data. The successful balance achieved through full logit optimal matching provides a template for future disability labour market research.

From a policy perspective, the findings underscore the need for disability-specific safeguards within crisis response frameworks. As elaborated in the preceding policy recommendations section, these include automatic earnings stabilisation transfers triggered by macroeconomic shocks, mandatory disability employment and pay gap reporting for large employers, improved collection of disaggregated disability data in national longitudinal surveys, and the pre-drafting of disability annexes to all major crisis response plans. Without such targeted interventions, general economic stabilisation measures will likely continue to leave disabled workers behind.

Several limitations of this study warrant acknowledgment. The exclusion of individuals who acquired or lost disability status during the period of interest, while necessary for internal validity, means the findings generalise only to the stable disability population. The relatively small number of treated units (221 matched disabled individuals) limits statistical power for subgroup analyses (e.g., by disability type or severity). Additionally, the study captures only short- to medium-term effects (up to four years post-pandemic); longer-term labour market trajectories remain unobserved. Finally, the analysis relies on self-reported wage data, which may contain measurement error, although there is no reason to suspect differential reporting bias across groups.

Future research may be directed along several avenues. First, as longer-term panel data become available, researchers should examine whether the earnings penalty observed in 2020–2022 persists, narrows, or widens over a five- to ten-year horizon. Second, investigations into the mechanisms underlying the effect—such as

differential job loss rates, occupational downgrading, reduced working hours, or wage bargaining power erosion: would complement the reduced-form estimates presented here. Third, comparative studies across different national contexts with varying disability support systems, labour market regulations, and pandemic policy responses would help establish the generalisability of the Russian case. Fourth, future work could employ alternative matching methods (e.g., genetic matching or coarsened exact matching) and sensitivity analyses (e.g., Rosenbaum bounds) to further assess robustness to unobserved confounding. Finally, scenario-based forecasting approaches could be developed, combining demographic and disability prevalence projections with econometric models to simulate future crisis impacts and inform strategic planning by ministries of labour and social protection.

In summary, this study provides robust causal evidence that the COVID-19 pandemic exacerbated labour market segmentation based on disability status in Russia. Disabled workers bore a substantial and persistent earnings penalty relative to non-disabled workers, a finding that aligns with and extends Labor Market Segmentation Theory. The results carry clear implications for both crisis preparedness and routine labour market governance: disability-inclusive policy cannot be an afterthought but must be designed *ex ante* into economic structures. By documenting the magnitude and persistence of this disparity, the thesis contributes to a growing body of evidence that systemic shocks, if unaddressed, deepen rather than disrupt existing social inequalities.

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Appendix A. Used python code to compute TWFE.

```
# Library imports
import matplotlib.pyplot as plt
import numpy as np
import pandas as pd
import polars as pl
import seaborn as sns
import statsmodels.formula.api as smf
from linearmodels.panel import PanelOLS

# Data import
panel = pd.read_parquet("../data/working_data.parquet")

# Plot average wages year comparison
df_wages = (
    panel[["year", "has_disablity", "wages"]]
    .groupby(["year", "has_disablity"])
    .mean()
    .reset_index()
)
df_wages["year"] = pd.to_datetime(df_wages["year"], format="%Y")
df_wages["has_disablity"] = df_wages["has_disablity"].map({0: "No", 1: "Yes"})
df_wages = df_wages.pivot(index="year", columns="has_disablity",
values="wages")

colors = {"Yes": "steelblue", "No": "lightblue"}
fig, ax = plt.subplots(figsize=(12, 5))

for column in df_wages.columns:
    ax.plot(
        df_wages.index,
        df_wages[column],
        marker="o",
        linewidth=2.5,
        markersize=8,
        color=colors.get(column, "#3498db"),
        label=f"Has Disability: {column}",
        markeredgecolor="white",
        markeredgewidth=1.5,
        alpha=0.9,
    )

ax.set_title(
    f"average Wages by year comparison",
    fontsize=16,
    fontweight="bold",
    pad=20,
    # fontfamily='serif'
)

ax.axvline(
    x=pd.Timestamp("2019-01-01"),
    color="gray",
    linestyle="--",
    linewidth=1.5,
    label="period of interest",
```

```

        alpha=0.7,
    )

    ax.axvline(
        x=pd.Timestamp("2020-01-01"), color="gray", linestyle="--",
        linewidth=1.5, alpha=0.7
    )
    ax.set_xlabel("year", fontsize=12, fontweight="semibold", labelpad=10)
    ax.set_ylabel(f"average Wages", fontsize=12, fontweight="semibold",
        labelpad=10)

    legend = ax.legend(
        title="Has Disability",
        title_fontsize=12,
        fontsize=11,
        loc="best",
        frameon=True,
        fancybox=True,
        shadow=True,
        edgecolor="#cccccc",
        facecolor="white",
    )
    legend.get_title().set_fontweight("bold")

    # Data manipulation
    panel_2019_2020 = panel.copy()
    panel_2019_2020["year"] = pd.to_datetime(panel_2019_2020["year"],
        format="%Y")
    panel_2019_2020_ids = panel_2019_2020["idind"]
    panel_2019_2020 = panel_2019_2020.set_index(["idind", "year"])

    # Serial correlation test computation
    fixed_effects_ind_model = PanelOLS.from_formula(
        "wages ~ period + EntityEffects", data=panel_2019_2020
    ).fit()
    print(fixed_effects_ind_model.summary)

    fe_resid = fe_model.resids

    resid_df = pd.DataFrame(
        {
            "idind": panel_2019_2020.reset_index()["idind"].values,
            "year": panel_2019_2020.reset_index()["year"].values,
            "resid": fe_resid.values,
        }
    ).sort_values(["idind", "year"])

    resid_df["resid_diff"] = resid_df.groupby("idind")["resid"].diff()
    resid_df["resid_diff_lag"] = resid_df.groupby("idind")["resid_diff"].shift(1)
    resid_df = resid_df.dropna()

    model_ols = smf.ols("resid_diff ~ resid_diff_lag",
        resid_df).fit(cov_type="HC1")
    print(model_ols.summary2())

```

```

# Difference in differences TWFE model computation
model_twfe = PanelOLS.from_formula(
    "wages ~ I(has_disability * period) + EntityEffects + TimeEffects",
    data=panel_2019_2020,
).fit(cov_type="clustered", cluster_entity=True, group_debias=False)
print(model_twfe.summary)

# Dynamic difference in differences TWFE model computation
panel = pl.read_parquet("../data/working_data_multiple_periods.parquet")
panel_2014_2024 = (
    panel.filter(
        (pl.col("period_relevance") > -5) & (pl.col("period_relevance") < 5)
    ).with_columns(pl.col("year").cast(pl.Int32).cast(pl.String).str.to_date("%Y"))
).to_pandas()
panel_2014_2024_dta = panel_2014_2024.set_index(["idind", "year"])

event_study_terms = []
for p in sorted(panel_2014_2024_dta["period_relevance"].unique()):
    if p != 0:
        col_name = f"rel_period_{p}_treated"
        panel_2014_2024_dta[col_name] = (
            (panel_2014_2024_dta["period_relevance"].values == p)
            & (panel_2014_2024_dta["has_disability_period"].values == 1)
        ).astype(int)
        event_study_terms.append(col_name)

X_event = panel_2014_2024_dta[event_study_terms]
y_event = panel_2014_2024_dta["wages"]

event_study_model = PanelOLS(
    y_event, X_event, entity_effects=True, check_rank=True
).fit(cov_type="clustered", cluster_entity=True)
print(event_study_model.summary)

# Dynamic difference in differences TWFE model plot
def plot_event_margins(model, measure_unit="RUB"):
    coef_names = [
        col for col in model.params.index if
        col.startswith("period_relevance")
    ]
    coefs = model.params[coef_names]
    std_errors = model.std_errors[coef_names]

    rel_periods = [int(re.search(r"-?\d+", name).group()) for name in
        coef_names]
    coef_df = pd.DataFrame(
        {
            "relative_period": rel_periods,
            "estimate": coefs.values,
            "std_error": std_errors.values,
        }
    )

    coef_df["conf_low"] = coef_df["estimate"] - 1.96 * coef_df["std_error"]

```

```

coef_df["conf_high"] = coef_df["estimate"] + 1.96 * coef_df["std_error"]

coef_df["type"] = np.where(
    coef_df["relative_period"] < 0, "Pre-treatment", "Post-treatment"
)

ref_row = pd.DataFrame(
    {
        "relative_period": [-1],
        "estimate": [0],
        "std_error": [0],
        "conf_low": [0],
        "conf_high": [0],
        "type": ["Pre-treatment"],
    }
)

coef_df = pd.concat([coef_df, ref_row], ignore_index=True)
coef_df = coef_df.sort_values("relative_period").reset_index(drop=True)

max_effect_row = coef_df.loc[coef_df["estimate"].idxmax()]
max_conf_high = coef_df["conf_high"].max()
min_conf_low = coef_df["conf_low"].min()

fig, ax = plt.subplots(figsize=(12, 7))
ax.axvspan(-6.5, 0, alpha=0.15, color=book_colors["light_gray"],
zorder=0)
ax.axvspan(3, 5, alpha=0.15, color=book_colors["light_gray"], zorder=0)
ax.axhline(
    y=0, linestyle="--", color=book_colors["muted"], linewidth=0.8,
alpha=0.8
)
ax.axvline(
    x=0, linestyle="--", color=book_colors["muted"], linewidth=0.8,
alpha=0.8
)
ax.axvline(
    x=3, linestyle="--", color=book_colors["muted"], linewidth=0.8,
alpha=0.8
)
ax.errorbar(
    coef_df["relative_period"],
    coef_df["estimate"],
    yerr=[
        coef_df["estimate"] - coef_df["conf_low"],
        coef_df["conf_high"] - coef_df["estimate"],
    ],
    fmt="o",
    color=book_colors["primary"],
    markersize=6,
    capsize=4,
    capthick=1.5,
    linewidth=0.8,
    ecolor=book_colors["primary"],
    elinewidth=0.8,
    zorder=3,
)

```

```

        annotation_y = max_conf_high * 0.7 if max_conf_high > 0 else
max_conf_high * 0.3
        ax.annotate(
            "pre-pandemic",
            xy=(-1.5, annotation_y),
            fontsize=13,
            color=book_colors["muted"],
            ha="center",
            style="italic",
        )
        ax.annotate(
            "pandemic",
            xy=(1.5, annotation_y),
            fontsize=13,
            color=book_colors["primary"],
            ha="center",
            style="italic",
        )
        ax.annotate(
            "post-pandemic",
            xy=(4.5, annotation_y),
            fontsize=13,
            color=book_colors["muted"],
            ha="center",
            style="italic",
        )
        ax.set_xlabel(
            "period relative to the event", fontsize=14,
color=book_colors["dark_gray"]
        )
        ax.set_ylabel(
            f"estimated effect ({measure_unit})",
            fontsize=14,
            color=book_colors["dark_gray"],
        )
        ax.set_title(
            "effect of COVID-19 on earnings over time",
            fontsize=16,
            fontweight="bold",
            color="#333333",
            pad=15,
        )
        ax.text(
            0.5,
            1.02,
            "period relevance k = -1 (2019)",
            transform=ax.transAxes,
            fontsize=11,
            color="grey",
            ha="center",
        )
        ax.set_xticks(range(-5, 5))
        ax.set_xlim(-6.5, 5)
        y_padding = (
            max_conf_high - min_conf_low) * 0.2 if max_conf_high != min_conf_low
else 1
    )

```

```

ax.set_ylim(min_conf_low - y_padding, max_conf_high + y_padding)
ax.spines["top"].set_visible(False)
ax.spines["right"].set_visible(False)
ax.spines["left"].set_color(book_colors["muted"])
ax.spines["bottom"].set_color(book_colors["muted"])
ax.tick_params(colors=book_colors["dark_gray"], labelsiz=11)
ax.grid(True, alpha=0.3, linestyle="--", linewidth=0.5)
plt.tight_layout()
plt.subplots_adjust(bottom=0.12)
plt.show()

return fig, ax, coef_df

```

Appendix B. R code used in PSM

```
# Library imports
library(MatchIt)
library(fixest)
library(stargazer)
library(modelsummary)
library(arrow)
library(MASS)
library(e1071)

# Data import
panel = read_parquet('../data/working_data.parquet')

# Box-Cox transformation
hist(panel$age, main = "Original Data", col = "lightblue")
qqnorm(panel$age, main = "Q-Q Plot (Original)")
qqline(panel$age, col = "red")

boxcox_result <- boxcox(lm(panel$age ~ 1), lambda = seq(-2, 2, 0.1))

best_lambda <- boxcox_result$x[which.max(boxcox_result$y)]
print(paste("Optimal lambda:", round(best_lambda, 3)))

transformed_data <- (panel$age^best_lambda - 1) / best_lambda

hist(transformed_data, main = "Transformed ( $\lambda = 0.263$ )", col = "lightgreen")
qqnorm(transformed_data, main = "Q-Q Plot (Transformed)")
qqline(transformed_data, col = "blue")
boxplot(transformed_data, main = "Boxplot (Transformed)", col = "lightgreen")

panel$age_box_cox = (panel$age^best_lambda - 1) / best_lambda

# PSM
psm_nn <- matchit(has_disablity ~ age_box_cox
                  + educ_level_university + is_female +
                  is_married + harmful_job + is_employed,
                  data = panel,
                  s.weights = ~ inwgt
                  )
print(summary(psm_nn, un = FALSE))

psm_full_logit <- matchit(has_disablity ~ age_box_cox
                        + educ_level_university + is_female +
                        is_married + harmful_job + is_employed,
                        data = panel,
                        s.weights = ~ inwgt,
                        method = "full",
                        link = "logit",
                        distance = "glm",
                        #exact = ~log_age
                        )
print(summary(psm_full_logit, un = FALSE))

# Compare PSM algorithms
love.plot(psm_full_probit, stats = c("m", "ks"), poly = 2, abs = TRUE,
          weights = list(nn = psm_nn),
```



```
drop.distance = TRUE, thresholds = c(m = .1),
var.order = "unadjusted", binary = "std",
shapes = c("circle filled", "triangle", "square"),
colors = c("lightcoral", "steelblue", "lightblue"),
sample.names = c("Original", "Full Matching", "NN Matching"),
position = "bottom")

panel_matched <- match_data(psm_full_logit)

write_parquet(panel_matched, '../data/working_data_matched.parquet')
```